Year 12 Maths - Polynomials

Similar questions - Stationary points and the number of real roots

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A curve has equation $y = 2x^3 - 9x^2 + 12x$.

(a) Find the coordinates of the stationary points of the curve. **[4]**

(b) Find the set of values of k for which the equation $2x^3 - 9x^2 + 12x = k$ has exactly one real root. **[2]**

(c) Find the values of k for which the equation has exactly two real roots. **[2]**

2 A box with no lid is made from a square sheet of card of side 24 cm by cutting a square of side x cm from each corner and folding up the sides.

(a) Show that the volume of the box is $V = 4x^3 - 96x^2 + 576x$ cm^3. **[2]**

(b) Find the value of x that gives the maximum volume, and find this volume. **[4]**

(c) Explain why the equation $V = 500$ has exactly two solutions for $0 < x < 12$. **[2]**

Question 1

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 6x^2 - 18x + 12 = 6(x-1)(x-2) = 0$, so $x = 1$ or 2. The stationary points are $(1,5)$ and $(2,4)$

(b) A positive cubic has a local maximum at $(1,5)$ and a local minimum at $(2,4)$. The line $y = k$ meets the curve once when $k < 4$ or $k > 5$

(c) Exactly two real roots when the line touches a turning point: $k = 4$ or $k = 5$

Question 2

(a) Base is $(24-2x)$ by $(24-2x)$ and height x: $V = x(24-2x)^2 = x(576-96x+4x^2) = 4x^3 - 96x^2 + 576x$

(b) $V' = 12x^2 - 192x + 576 = 12(x-4)(x-12) = 0$. For $0 < x < 12$ the stationary point is $x = 4$, giving $V = 4(16)^2 = 1024$ cm^3 (the maximum, as $V = 0$ at $x = 0$ and $x = 12$)

(c) $V = 0$ at $x = 0$, rises to the maximum 1024 at $x = 4$, then falls to 0 at $x = 12$. The line $V = 500$ lies between 0 and 1024, so it crosses the curve once on each side of the maximum